Data Structures
CSE-207

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Array

- Data Structure can be classified as:
  - Linear
  - non-linear
- Linear (elements arranged in sequential in memory location) i.e. array & linear link-list
- Non-linear such as a tree and graph.
- Operations:
  - Traversing, Searching, Inserting, Deleting, Sorting, Merging
- Array is used to store a fix size for data and a link-list the data can be varies in size.

Array

- Linear array
  - A linear array is a list of a finite number of homogeneous data elements such that:
    - The elements are referenced by an index set.
    - Elements are stored in successive memory locations.
  - The number n of element is called length or size of the array
- How we calculate the size?
  - Length=UB-LB+1
    - UB=largest index called upper bound
    - LB=smallest index called lower bound

Array

- How we calculate the size?
  - Suppose a company uses an array AUTO to record the number of automobile sold by each year from 1932 to 1984.
    - AUTO[K] is the no. of automobile sold in year K
    - Let the index start with 1932
    - So LB= 1932 and UB=1984
    - So length=UB-LB+1

      =1984-1932+1

      =55

Inserting an Element

- Inserting element at the end is easy.
- Inserting in middle is difficult.
- Half of the elements need to move downward!!!

We have an array LA

K=3, N=5, K<=N,

ITEM=Inserting Element=15

We want to Insert ITEM kth Position.

10
25
35
68
78

So Let do it…..

Initial
K=3, N=5

10
25
35
68
78

1. Set J:=N
2. Repeat Step 3 and 4 While J>=K
3. Set LA[J+1]:=LA[J]
4. Set J:=J-1
5. Set LA[K]:=ITEM
6. Set N=N+1
7. Exit.

J=5

So Let do it…..

10
25
35
68
78

Initial
K=3, N=5

1. Set J:=N
2. Repeat Step 3 and 4 While J>=K
3. Set LA[J+1]:=LA[J]
4. Set J:=J-1
5. Set LA[K]:=ITEM
6. Set N=N+1
7. Exit.

J=5
LA[6]=LA[5]
J=4

So Let do it…..

10
25
35
68
78

Initial
K=3, N=5

1. Set J:=N
2. Repeat Step 3 and 4 While J>=K
3. Set LA[J+1]:=LA[J]
4. Set J:=J-1
5. Set LA[K]:=ITEM
6. Set N=N+1
7. Exit.

J=4
LA[5]=LA[4]
J=3

So Let do it…..

Initial
K=3, N=5

10
25
35
68
78

1. Set J:=N
2. Repeat Step 3 and 4 While J>=K
3. Set LA[J+1]:=LA[J]
4. Set J:=J-1
5. Set LA[K]:=ITEM
6. Set N=N+1
7. Exit.

J=3
LA[4]=LA[3]
J=2

So Let do it…..

Initial
K=3, N=5

10
25
15
35
68
78

1. Set J:=N
2. Repeat Step 3 and 4 While J>=K
3. Set LA[J+1]:=LA[J]
4. Set J:=J-1
5. Set LA[K]:=ITEM
6. Set N=N+1
7. Exit.

LA[3]:=ITEM=15

Finally
N=6

Deleting an element

- Deleting an element from end location is not a difficult.
- Deleting from middle need to move upward one location for subsequence element
- We have an array LA

10
25
35
68
78
15

K=3, N=6, K<=N

We want to delete Kth Element.

So Let do it…..

10
25
68
78
15

Initial
K=3, N=6

1. Set ITEM=LA[k]
2. Repeat for J= K to N-1
   Set LA[J]=LA[J+1]
3. Set N=N-1
4. Exit.

1st: J=K=3 So LA[3]=LA[4]
2nd: J=K=4 So LA[4]=LA[5]
3rd: J=K=5 So LA[5]=LA[6]

Finally
N=5

END